

Abinit ground-state performances on CPU

Abinit Developer Workshop 2026

Lucas Baguet^{1,2}, Marc Torrent^{1,2}

1 : CEA, DAM, DIF, F-91297, Arpajon, France

2 : Université Paris-Saclay, CEA, Laboratoire Matière en Conditions Extrêmes, 91680 Bruyères le Châtel, France

8th of June 2026

Table of contents

Algorithms and parallelization schemes

Computation time for Fe supercells

Conclusion

Table of contents

Algorithms and parallelization schemes

Computation time for Fe supercells

Conclusion

A bit of Abinit History : algorithms (date of publications)

- 2002 : Conjugate Gradient (CG) - $w_{optalg} = 0/10$
X. Gonze et al, Computational Materials Science, Volume 25, Issue 3, 2002
- 2008 : LOBPCG - $w_{optalg} = 4/14$
F. Bottin et al. Computational Materials Science, Volume 42, Issue 2, 2008
- 2015 : Chebyshev Filtering (Cheb.Fi.) - $w_{optalg} = 1$
A. Levitt, M. Torrent, Computer Physics Communications, Volume 187, 2015
- 2020 : LOBPCG 2, compatible with OpenMP+vecto - $w_{optalg} = 114$
X. Gonze et al, Computer Physics Communications, Volume 248, 2020
- 2025 : RMM-DIIS (with CG or LOBPCG) - $rmm_diis > 0$
M. Verstraete et al, The Journal of Chemical Physics, Volume 163, 2025
- 2025 : Cheb.Fi. 2, compatible with OpenMP+vecto - $w_{optalg} = 111$
M. Verstraete et al, The Journal of Chemical Physics, Volume 163, 2025
- 2025 : LOBCPG+Cheb.Fi. 3 - $w_{optalg} = 114/111 + cprj_in_memory = 1$
M. Verstraete et al, The Journal of Chemical Physics, Volume 163, 2025

A bit of Abinit History : parallelism

- 2002 : MPI on k-points (*np_spkpt*)
- 2008 : MPI on band (*npband*) and FFT (*npfft*)
- 2016 : GPU (*gpu_option* = 1 or *gpu_option* = *GPU_LEGACY*)
- 2020 : Efficient OpenMP + vectorization (*export OMP_NUM_THREADS N*)
- 2025 : GPU (*gpu_option* = 2 or *gpu_option* = *GPU_OPENMP*)

<i>wfoptalg</i>	CG	LOBPCG		Cheb.Fi.		RMM-DIIS	<i>cprj_in_memory</i> = 1	
	0/10	4/14	114	1	111	<i>rmm_diis</i> > 0	114	111
<i>np_spkpt</i>	X	X	X	X	X	X	X	X
<i>npband</i>		X	X	X	X	X	X	X
<i>npfft</i>		X	X	X		X		
<i>threads</i>			X		X	X	X	X
<i>gpu_option</i> = 2			X		X	X		

Algorithm parameters

Conjugate Gradient : number of iterations $nline$

$block\ size = 1$

$number\ of\ blocks = nband$

LOBPCG : $nline +$ number of blocks $nblock_lobpcg$ (or set indirectly by $bandpp$ and $npband$)

$block\ size = nband / nblock_lobpcg = npband * bandpp$

$1 \leq nblock_lobpcg \leq nband$

Chebyshev Filtering : $mdeg_filter$ (or $nline$)

$block\ size = nband$

$number\ of\ blocks = 1$

RMM-DIIS : rmm_diis sets the number of SCF iterations done with CG or LOBPCG

$block\ size = nband$

$number\ of\ blocks = 1$



How to optimize ground state computations (CPUs only) ?

- Algorithms : choice among 8 possibilities
- Alg. parameters : 1 value to set (or 2 for LOBPCG)
- Parallelization : one has to distribute N cores among 3 levels

How to choose the best combination ?

How to optimize ground state computations (CPUs only) ?

- Algorithms : choice among 8 possibilities
- Alg. parameters : 1 value to set (or 2 for LOBPCG)
- Parallelization : one has to distribute N cores among 3 levels

How to choose the best combination ?

Quick solution : *autoparal* = 1. However :

- does not select the algorithms and their parameters
- does not take into account local configuration (hardware+software)
- works with a fixed number of threads
- **not updated for a while**

How to optimize ground state computations (CPUs only) ?

- Algorithms : choice among 8 possibilities
- Alg. parameters : 1 value to set (or 2 for LOBPCG)
- Parallelization : one has to distribute N cores among 3 levels

How to choose the best combination ?

Quick solution : *autoparal* = 1. However :

- does not select the algorithms and their parameters
- does not take into account local configuration (hardware+software)
- works with a fixed number of threads
- **not updated for a while**

We need more data. Let's try a systematic exploration of Abinit performances

There are **many** configurations to test. Here we will focus on (arbitrary choice) :

- LOBPCG and Chev.Fi., with or without *cprj_in_memory*
- Paral. : *np_spkt*, *npband*, threads



Table of contents

Algorithms and parallelization schemes

Computation time for Fe supercells

Conclusion

Hardware+software setup

Hardware :

- CPUs : 128 cores per node ($2 \times$ AMD Milan EPYC 7763 64C 2.45GHz)
- Memory : 256 GB of RAM per node (= 2GB/core)
- Network : Atos BXI v2 100 Gb/s

Software :

- gcc 12.3.0
- openmpi 4.1.7
- mkl 25.0.0 (so multi-threaded BLAS/LAPACK)
- abinit 10.6.7

Test cases

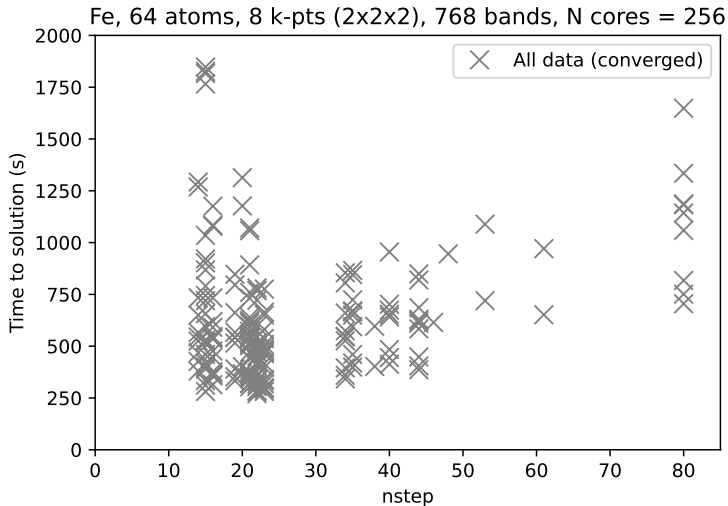
Supercells of Fe :

- PAW pseudo : JTH-v2.0-GGA-PBE
- 12 bands per atom, $tsmear = 300K$
- magnetic system : $nsppol = 2$
- "MD-like" configuration : $nsym = 1, optforces = 1$ (but ions not moved : first MD step only)

System sizes :

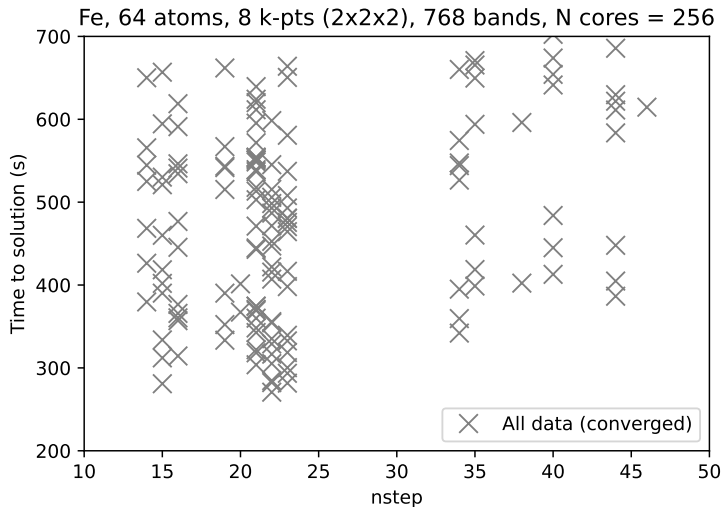
- 8 atoms ($2 \times 2 \times 2$ cells), 36 k-points ($4 \times 4 \times 4$), 96 bands, $tolvrs = 1e - 6$
- **64 atoms ($4 \times 4 \times 4$ cells), 8 k-points ($2 \times 2 \times 2$), 768 bands**, $tolvrs = 1e - 4$
- 512 atoms ($8 \times 8 \times 8$ cells), 1 k-points, 6144 bands, $tolvrs = 1e - 2$

Fe : 64 atoms, 256 cores



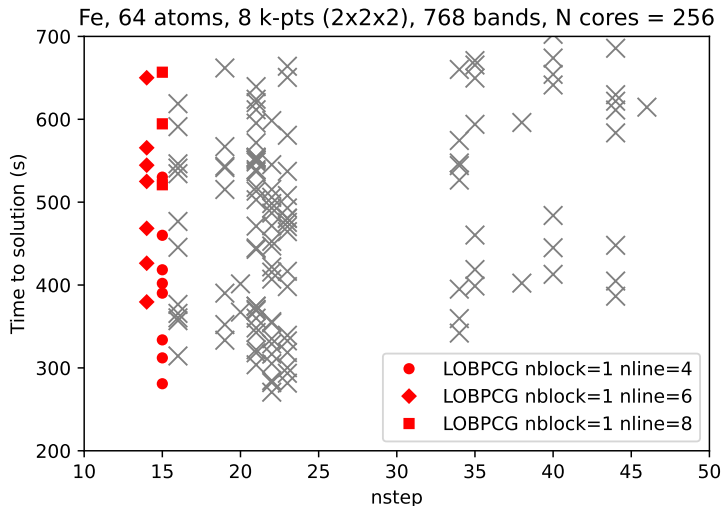
param.	values
<i>wfoptalg</i>	114,111
<i>nblock</i>	1,2,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom)



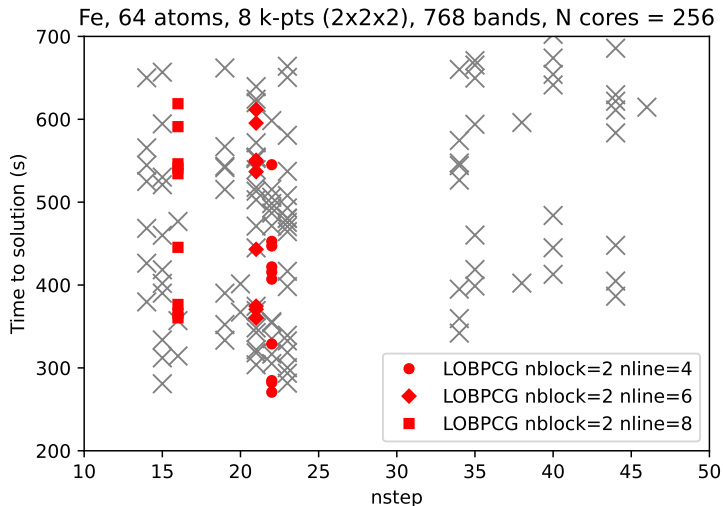
param.	values
<i>wfoptalg</i>	114,111
<i>nblock</i>	1,2,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 1 block



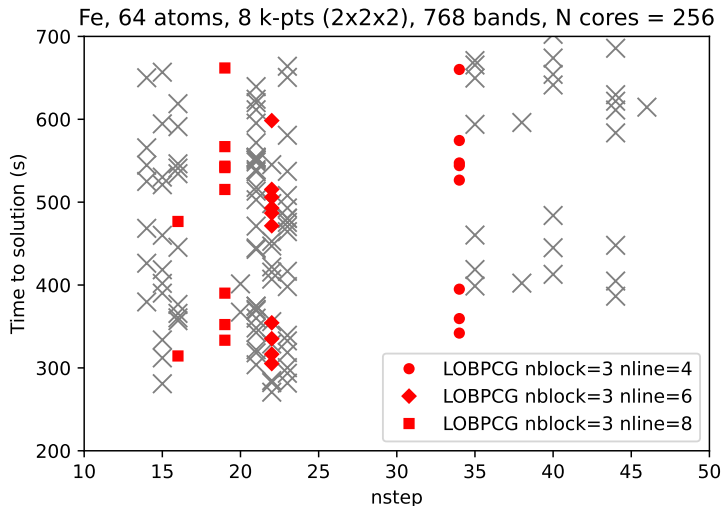
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1 ,2,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 2 blocks



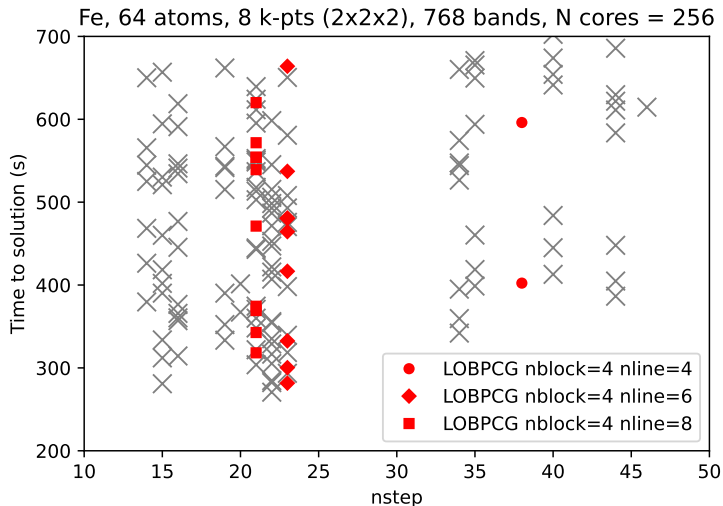
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1, 2 ,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 3 blocks



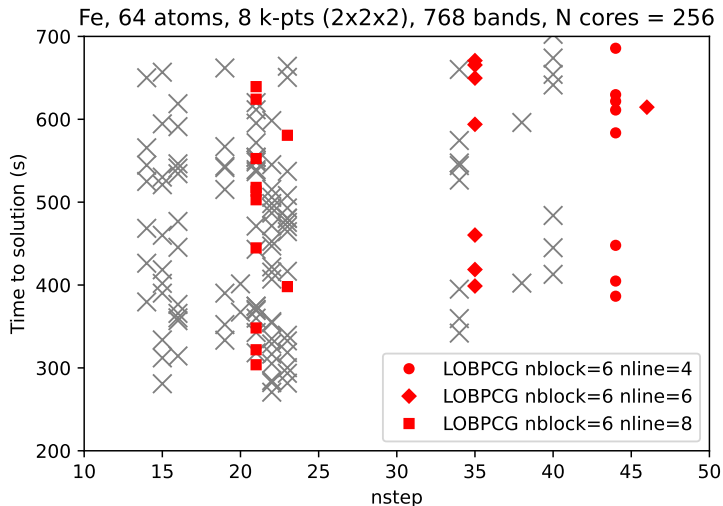
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1,2, 3 ,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 4 blocks



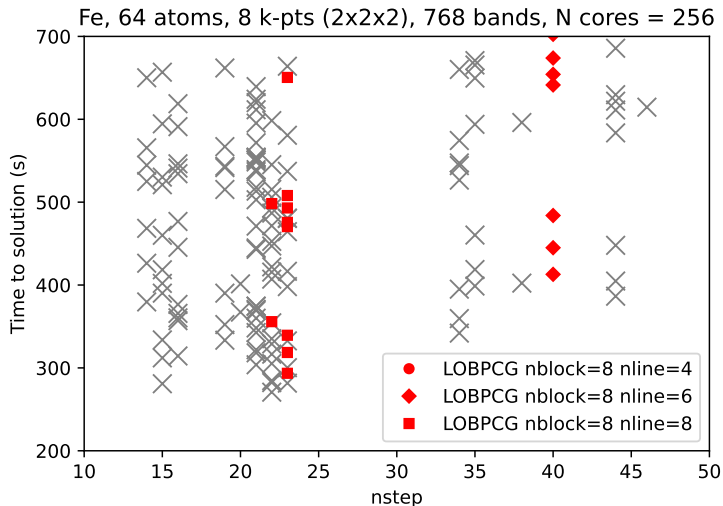
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1,2,3, 4 ,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 6 blocks



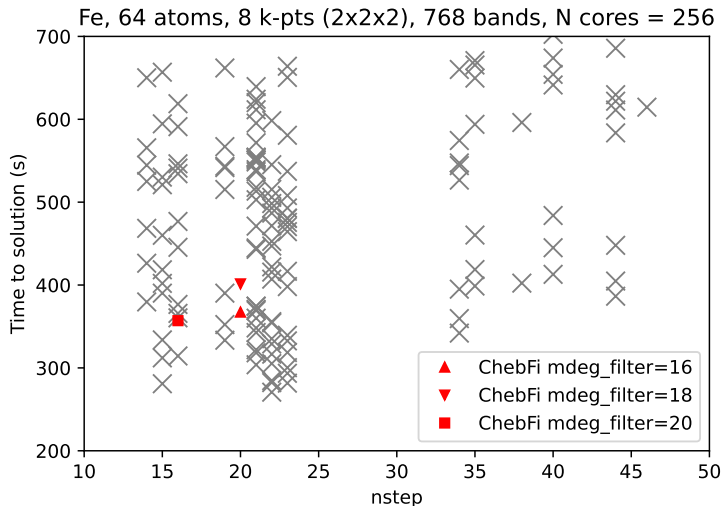
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1,2,3,4, 6 ,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - LOBPCG 8 blocks



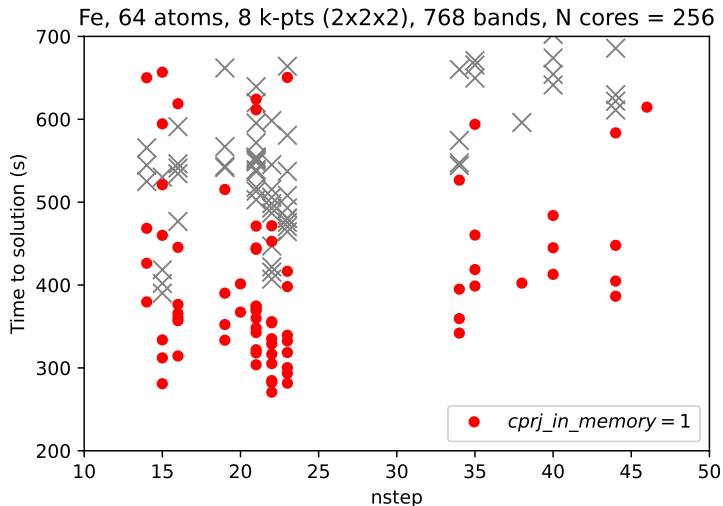
param.	values
<i>wfoptalg</i>	114 ,111
<i>nblock</i>	1,2,3,4,6, 8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - Chev.Fi.



param.	values
<i>wfoptalg</i>	114, 111
<i>nblock</i>	1,2,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores (zoom) - *cprj_in_memory*



param.	values
<i>wfoptalg</i>	114,111
<i>nblock</i>	1,2,3,4,6,8
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	8
<i>npband</i>	1,2,4,8,16,32
<i>nthreads</i>	1,2,4,8,16,32

Fe : 64 atoms, 256 cores - Top 15

rank	wfoptalg	cprj.	nblock	nline	npkpt	npband	nthreads	nstep	time (s)
1	114	1	2	4	8	4	8	22	271
2	114	1	1	4	8	1	32	15	281
3	114	1	4	6	8	4	8	23	282
4	114	1	2	4	8	2	16	22	282
5	114	1	2	4	8	1	32	22	285
6	114	1	8	8	8	4	8	23	294
7	114	1	4	6	8	2	16	23	300
8	114	1	6	8	8	4	8	21	304
9	114	1	3	6	8	4	8	22	305
10	114	1	1	4	8	2	16	15	312
11	114	1	3	8	8	1	32	16	314
12	114	1	3	6	8	2	16	22	317
13	114	1	4	8	8	4	8	21	318
14	114	1	8	8	8	2	16	23	319
15	114	1	6	8	8	2	16	21	322

What we can conclude :

- LOBPCG gives best results
- $cprj_in_memory = 1$ gives best results

1st config. with $cprj_in_memory = 0$:
 42^{th} : $time = 390s$

1st config. with Cheb.Fi :
 28^{th} : $time = 357s$

However :

- many combinations of $nblock$, $nline$, $npband$, $nthreads$ give similar results

Fe : 512 atoms, 1024 cores - Top 15

rank	wfoptalg	cprj.	nblock	nline	npkpt	npband	nthreads	nstep	time
1	114	1	6	4	1	16	64	23	6668
2	114	1	6	4	1	32	32	23	6863
3	114	1	6	8	1	16	64	14	7587
4	114	1	6	8	1	32	32	14	7793
5	111	1	1	18	1	32	32	35	8035
6	114	1	4	6	1	16	64	15	8073
7	114	1	4	6	1	32	32	15	8219
8	114	1	4	4	1	16	64	23	8316
9	111	1	1	18	1	16	64	35	8457
10	114	1	4	4	1	32	32	23	8465
11	114	1	6	4	1	64	16	23	8929
12	114	1	3	6	1	16	64	13	9088
13	114	1	3	4	1	16	64	20	9122
14	114	1	6	6	1	16	64	23	9170
15	114	1	3	4	1	32	32	20	9275

param.	values
<i>wfoptalg</i>	114,111
<i>nblock</i>	1,2,3,4,6
<i>nline</i>	4,6,8
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	1
<i>np_spkpt</i>	1
<i>npband</i>	16,32,64
<i>nthreads</i>	16,32,64

■ LOBPCG gives best results

1st config. with Chev.Fi :

5th : *time* = 8035s

Fe : 8 atoms, 12 cores - Top 15

rank	wfoptalg	cprj.	nblock	nline	npkpt	npband	nthreads	nstep	time
1	114	1	4	6	12	1	1	23	153
2	114	1	2	6	12	1	1	21	169
3	114	1	4	6	6	1	2	23	170
4	114	1	3	6	6	2	1	21	171
5	111	1	1	6	4	3	1	42	178
6	111	1	1	6	6	2	1	42	178
7	114	1	4	6	6	2	1	24	181
8	114	1	3	6	3	2	2	21	188
9	114	1	4	6	4	3	1	24	188
10	114	1	4	6	3	1	4	23	194
11	111	1	1	6	3	2	2	42	194
12	114	1	2	6	6	1	2	21	194
13	114	1	3	6	12	1	1	27	194
14	114	1	2	6	6	2	1	23	196
15	114	1	1	6	12	1	1	17	196

param.	values
<i>wfoptalg</i>	114,111
<i>nblock</i>	1,2,3,4,6
<i>nline</i>	6
<i>mdeg_filter</i>	16,18,20
<i>cprj.</i>	0,1
<i>np_spkpt</i>	1,2,3,4,6,12
<i>npband</i>	1
<i>nthreads</i>	1,2,3,4,6,12

■ LOBPCG gives best results

1st config. with Chev.Fi :

5th : *time* = 178s



Table of contents

Algorithms and parallelization schemes

Computation time for Fe supercells

Conclusion



Conclusion

Reminder : this is **very small** study :

- only one hardware+software configuration
- only one physical system (with different sizes)
- only one number of cores (per system size)

Conclusion

Reminder : this is **very small** study :

- only one hardware+software configuration
- only one physical system (with different sizes)
- only one number of cores (per system size)

In this context, what did we learn ?

- LOBPCG+*cprj_in_memory* = 1 seems to be a good starting choice
- It is very difficult to tune LOBPCG : *nblock*, *nline*, *npband*, *nthreads* are correlated
- Cheb.Fi.+*cprj_in_memory* = 1 can be interesting even for "small" systems

Conclusion

Reminder : this is **very small** study :

- only one hardware+software configuration
- only one physical system (with different sizes)
- only one number of cores (per system size)

In this context, what did we learn ?

- LOBPCG+*cprj_in_memory* = 1 seems to be a good starting choice
- It is very difficult to tune LOBPCG : *nblock*, *nline*, *npband*, *nthreads* are correlated
- Cheb.Fi.+*cprj_in_memory* = 1 can be interesting even for "small" systems

What's next ?

- Complete the database : try different number of cores ?
- Complete the database : try other systems ?
- Complete the database : try other algorithms ?
- Apply machine-learning methods on a local database to predict the optimal configuration ?